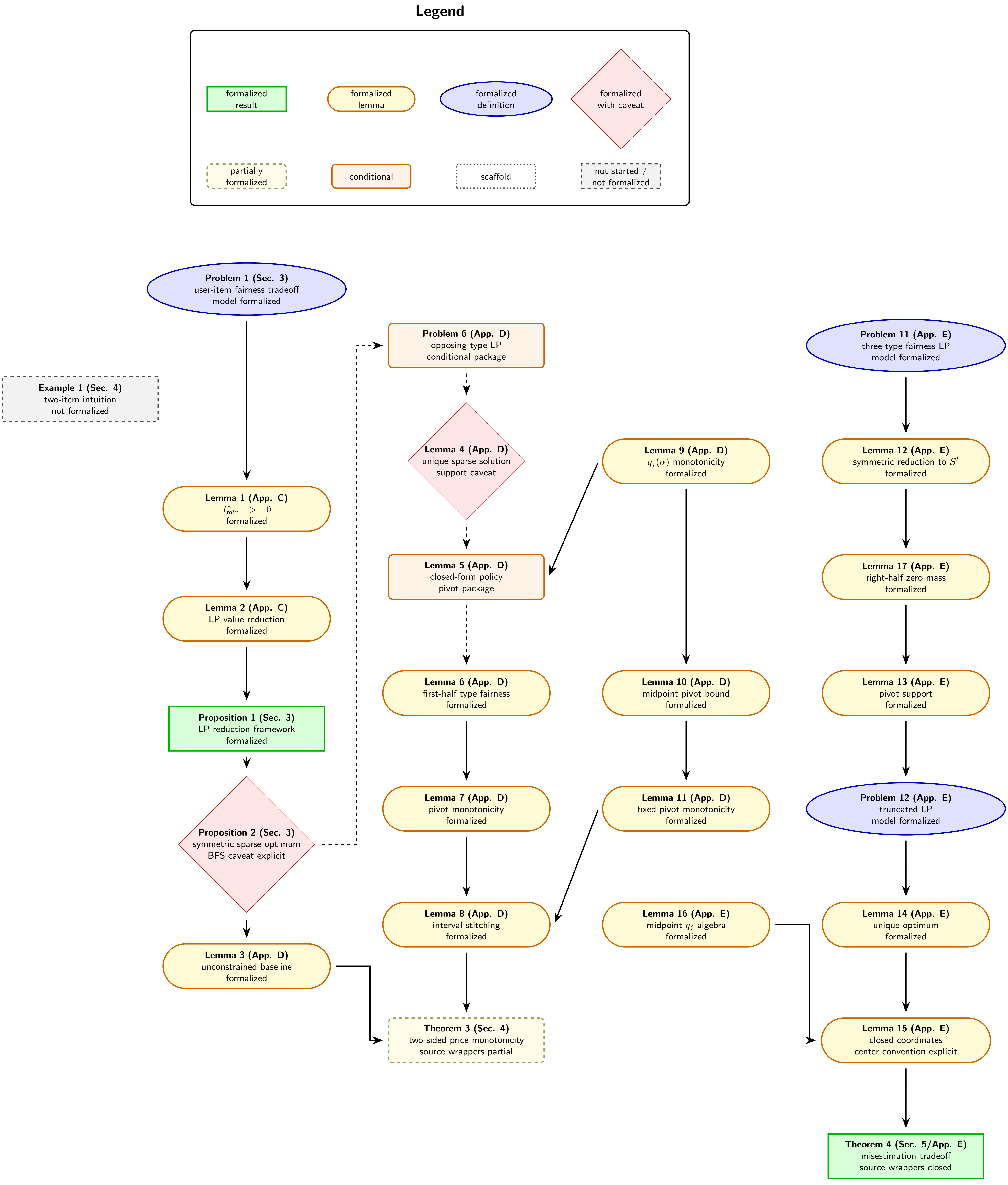
Legend
formalized result
formalized lemma
formalized definition
formalized with caveat
partially formalized
conditional
scaffold
not started / not formalized
Problem 1 (Sec. 3) user-item fairness tradeoff model formalized
Example 1 (Sec. 4) two-item intuition not formalized
Lemma 1 (App. C) $I^*_{\min} > 0$ formalized
Lemma 2 (App. C) LP value reduction formalized
Proposition 1 (Sec. 3) LP-reduction framework formalized
Proposition 2 (Sec. 3) symmetric sparse optimum BFS caveat explicit
Lemma 3 (App. D) unconstrained baseline formalized
Problem 6 (App. D) opposing-type LP conditional package
Lemma 4 (App. D) unique sparse solution support caveat
Lemma 5 (App. D) closed-form policy pivot package
Lemma 6 (App. D) first-half type fairness formalized
Lemma 7 (App. D) pivot monotonicity formalized
Lemma 8 (App. D) interval stitching formalized
Theorem 3 (Sec. 4) two-sided price monotonicity source wrappers partial
Lemma 9 (App. D) $q_j(\alpha)$ monotonicity formalized
Lemma 10 (App. D) midpoint pivot bound formalized
Lemma 11 (App. D) fixed-pivot monotonicity formalized
Lemma 16 (App. E) midpoint q_j algebra formalized
Problem 11 (App. E) three-type fairness LP model formalized
Lemma 12 (App. E) symmetric reduction to S' formalized
Lemma 17 (App. E) right-half zero mass formalized
Lemma 13 (App. E) pivot support formalized
Problem 12 (App. E) truncated LP model formalized
Lemma 14 (App. E) unique optimum formalized
Lemma 15 (App. E) closed coordinates center convention explicit
Theorem 4 (Sec. 5/App. E) misestimation tradeoff source wrappers closed